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Incorporation of two-dimensional correlation analysis into discriminant analysis as a potential tool for improving discrimination accuracy: Near-infrared spectroscopic discrimination of adulterated olive oils

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ABSTRACT

A strategy of combining temperature-induced spectral variation and two-dimensional correlation (2D-COS) analysis as a potential tool to improve accuracy of sample discrimination is suggested. The potential application of this method was evaluated using near-infrared (NIR) spectroscopic discrimination of adulterated olive oils. Rather than utilizing static spectral information at a certain temperature, dynamic spectral features induced by an external perturbation such as temperature change would be more informative for sample discrimination, and 2D-COS analysis was a reliable choice to characterize temperature-induced spectral variation. For evaluation, NIR spectra of 9 pure olive oils and 90 olive oils adulterated with canola, soybean, and corn oils (adulteration rate: 5%) were collected at four different temperatures (20, 27, 34, 41 °C). In constant-temperature measurements, the scores of pure and adulterated samples obtained by principal component analysis (PCA) were considerably overlapped. When 2D-COS analysis was performed using temperature-varied (20–41 °C) spectra and the resulting power spectra from 2D synchronous correlation spectra were used for PCA, identification of the two groups was noticeably enhanced and subsequent *k*-nearest neighbor (*k*-NN)-based discrimination accuracy substantially improved to 86.4%. While, the accuracies resulted in the constant-temperature measurements ranged only from 50.9 to 55.8%. The dynamic temperature-induced spectral variation of the samples effectively featured by 2D-COS analysis was ultimately more informative and allowed improvement in accuracy.

1. Introduction

Vibrational spectroscopy has been progressively studied as a fast analytical tool for identification of adulterated edible oils such as olive and sesame oils [1–12]. In these previous reports, chemometrics-based discriminant analysis has been mostly adopted to extract relevant spectral information from complexly overlapped sample spectra, since the molecular structures of components in edible oils (such as oleic, linoleic, and linolenic acids) are similar. Under this circumstance, one straightforward method to improve discrimination accuracy could be enhancement of spectral specificity (discriminability or selectivity) among individual components in a sample.

When the temperature of a mixed sample varied, the temperature-driven spectral variations of individual components became more diverse. Therefore, at a particular temperature, spectral discrimination among the components was enhanced, so improving accuracy of multivariate analysis. Using this strategy, optimal temperatures providing improved accuracies were identified in the vibrational spectroscopic determinations of component concentrations in synthetic hydrocarbon mixtures [13], density of polyethylene (PE) pellet [14,15], and viscosity of lube-based oil (LBO) [16,17]. The strategy was easy to execute by acquiring sample spectra at an optimal temperature; however, it was still based on recognition of static spectral information at a constant temperature.

Alternatively, dynamic spectral information induced by an external perturbation such as temperature change would be disparately characteristic and progression of temperature-driven structural variations of components are individually diverse. So, such a method would allow greater sample-to-sample discernibility, eventually leading to enhanced discrimination. For characterization of perturbation-induced spectral variation, two-dimensional correlation (2D-COS) analysis is a reliable choice, since it is able to capture unforeseen and minute spectral

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Table 1
Compositions of major fatty acids in olive, canola, corn, and soybean oils [28].

	Oleic Acid	Linoleic Acid	Linolenic Acid	Palmitic Acid	Stearic Acid
Olive oil	76%	7%	1%	12%	3%
Canola oil	56%	26%	10%	4%	2%
Corn oil	31%	52%	1%	13%	3%
Soybean oil	23%	51%	7%	10%	4%

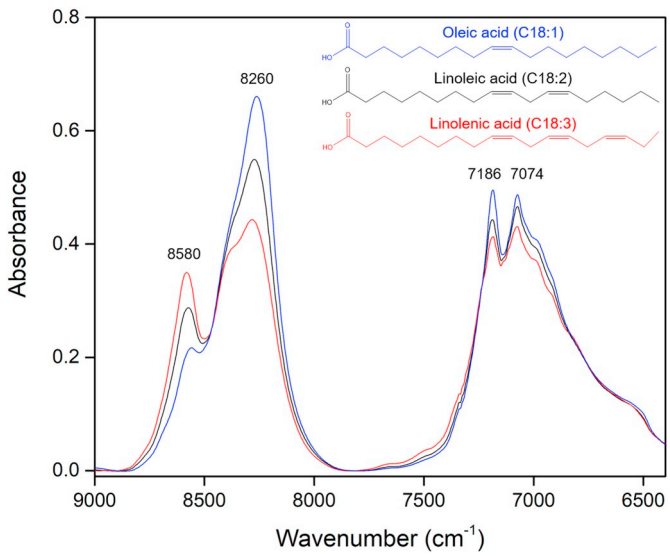

Fig. 1. NIR spectra of oleic, linoleic, and linolenic acids in the 9000-6400 cm^{-1} range. Molecular structures of these fatty acids are also shown.

variations such as shift of band position and change of band shape [18–24]. Thus, near-infrared (NIR) spectroscopic discrimination of *Fructus Lycii* (a Chinese herbal medicine) samples from four different geographic regions was attempted by analyzing temperature-varied sample spectra using 2D-COS analysis [25]. However, it was based on only qualitative observation of different 2D-COS patterns using a limited number of samples, without an attempt to provide tangible quantitative discrimination accuracy. Also, NIR or infrared (IR) spectroscopic discriminations of adulterated powdered milks by melamine and adulterated sesame oils by corn oil in conjunction with 2D-COS analysis were accomplished using the correlation patterns as input data [26,27] for diverse discriminant analyses such as kernel orthogonal projection to latent structure (K-OPLS) and multi-way partial least squares discriminant analysis (NPLS-DA). For 2D-COS analysis in these studies, two spectra (pure sample spectrum vs. pure sample spectrum or pure sample spectrum vs. adulterated sample spectrum) were used and the applied perturbation was the change of adulterant concentration, rather than temperature variation.

The major goal of this research was to demonstrate the utility for combining temperature-induced spectral variation and 2D-COS analysis as a potential tool to improve accuracy of sample discrimination. For this purpose, NIR spectroscopic identification of adulterated olive oils was attempted as a model study. Temperature-induced spectral variations between a pure oil and an adulterated oil are expected to be dissimilar, and their behaviors could be effectively distinguished using 2D-COS analysis, potentially providing more sample-discernible information for discriminant analysis. NIR spectra of 9 pure olive oils and 90 oils adulterated with canola, soybean, or corn oils were collected at four different temperatures (20, 27, 34, 41 °C). The adulteration rates of all the samples were low, 5%, to examine the potential of this strategy under a rather challenging condition. To understand the sample spectral features better, NIR spectra of major components in the oils (oleic, linoleic, and linolenic acids) were also collected, and their temperature-induced spectral features were examined.

Then, the distribution of principal component (PC) scores corresponding to pure and adulterated samples obtained from principal component analysis (PCA) was analyzed to examine the potential temperature dependency in the discrimination. Next, 2D-COS analysis was performed using the temperature-varied spectra, and potential differences in the 2D-COS patterns of pure and adulterated samples were examined. Power spectra obtained from the corresponding synchronous 2D correlation spectra were used for PCA, and subsequent score distribution was compared with those of constant-temperature measurements. To quantitatively assess discrimination accuracies, one-class *k*-nearest neighbor (*k*-NN) was employed. Finally, discrimination accuracies in the constant- and variable-temperature measurements were compared, and the utility of the proposed method for discriminant analysis was discussed.

2. Experimental section

Nine imported extra virgin olive oils (hereinafter referred to as olive oils) produced by different manufacturers were purchased. Also, three canola, four soybean, and three corn oils of different brands were obtained at local markets. To prepare an adulterated sample, contaminant oil was added into an olive oil to an adulteration rate of 5.0%, and this mixing was repeated without sample duplication. Therefore, a total of 90 adulterated olive oils (9 × 10 samples) were prepared along with 9 pure olive oils. Table 1 shows the compositions of fatty acids in olive, canola, corn, and soybean oils, obtained from the Food and Agricultural Organization of the United Nations (Codex Standards for Fats and Oils from Vegetable Sources) [28].

For spectral acquisition, each sample (1.8 mL) was transferred into a cylindrical glass vial (Wheaton Inc., NJ, USA) with an inner diameter of 1.0 cm, and the corresponding transmission NIR spectrum (9000–4000 cm^{-1} range) was acquired using an FT-NIR spectrometer (ABB) equipped with a quartz-halogen light source and an InGaAs detector. Sample temperature was controlled with a precision of ± 0.1 °C during measurement using a temperature controller (Watlow Inc., MO, USA). Perfluorohexane (C_6F_{14}) was used to produce a background absorbance spectrum, since its absorbance was close to zero in the measured spectral range. Each sample spectrum was collected in triplicate by 64 scans with 8 cm^{-1} resolution. All samples were measured at 20, 27, 34, and 41 °C. Baseline of each spectrum was corrected by using asymmetric least square (AsLS) before further analysis [29,30].

One-class *k*-NN, that determines whether or not a prediction sample belongs to a target class, was used since a single class classification was involved in this study [31,32]. In one-class *k*-NN, initially *K* sample (within a target class) closest to a prediction sample is determined and the average Euclidian distance $\overline{D_k}$ between these two samples is calculated. Next, *J* sample (also within the same target class) closest to *K* sample is found and the average Euclidian distance $\overline{D_j}$ between these two samples is calculated. Finally, using the threshold θ, if $\frac{\overline{D_j}}{\overline{D_k}} < \theta$, then a prediction sample is considered as a member of a target class. In this study, θ was set to 1.0, corresponding to equal weigh on both $\overline{D_k}$ and $\overline{D_j}$. In order to determine *K* and *J* samples, the numbers of nearest neighbors separately assigned to 1 or 2, then resulting accuracies in these four combinatorial cases were compared. PCA and one-class *k*-NN were performed using Matlab R2011a (Math Works Inc., USA).

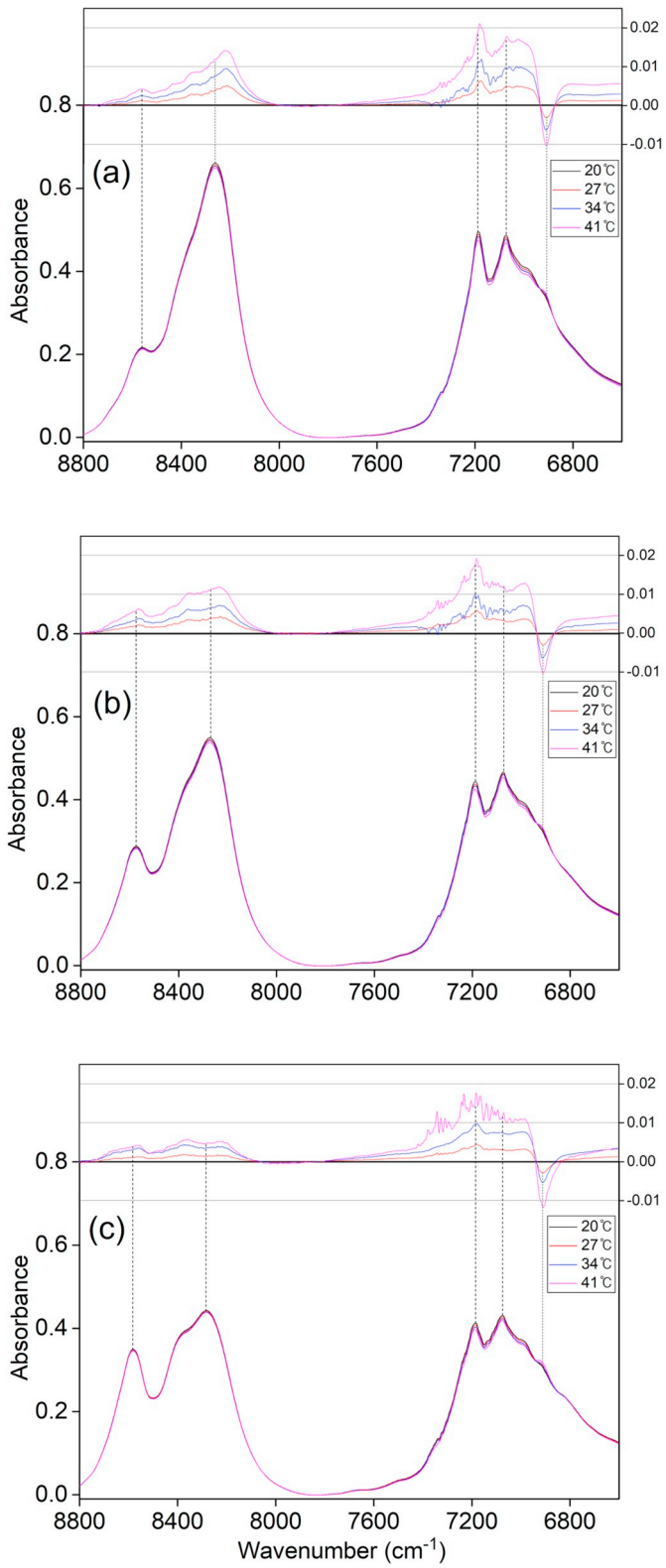

Fig. 2. NIR spectra of oleic (a), linoleic (b) and linolenic acids (c) measured at 20, 27, 34, and 41 °C. The difference spectra obtained by subtracting the spectra at 27, 34, 41 °C from the spectrum at 20 °C are displayed at the top.

3. Results and discussion

3.1. NIR spectral features of major fatty acids

Fig. 1 shows NIR spectra (9000-6400 cm^{-1} range) of oleic, linoleic, and linolenic acids (major pure components in edible oils) measured at 20 °C, with an inset for each molecular structure. Among these fatty acids, the numbers of contained double bonds (degrees of unsaturation) are different. The 8580 cm^{-1} band corresponding to the C=C bond clearly increases from oleic to linolenic acids, while the band of aliphatic chain around 8260 cm^{-1} decreases. The overall band shapes in the 7800-6400 cm^{-1} range, showing diverse overtone bands of CH_3 and methylene chain, are generally similar, while absorbance decreases from oleic to linolenic acids. Another noticeable trend is that the relative intensity between the 7186 and 7074 cm^{-1} bands varies with degree of unsaturation. Overall, NIR spectral features of the fatty acids apparently differ according to molecular structure; no peculiar bands specific to a given fatty acid were observed.

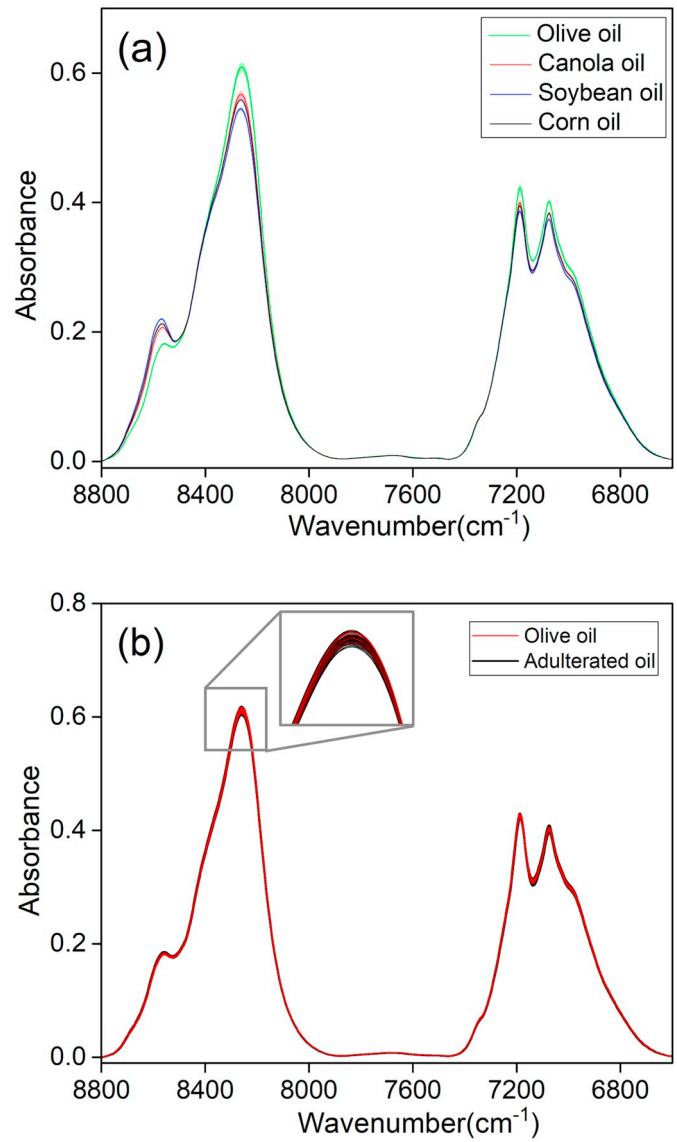

Fig. 3. Average NIR spectra of all pure olive, canola, soybean, and corn oils used in the study (a). The shades superimposed with the spectra designate the corresponding standard deviations, equivalent to sample-to-sample variation. (b) NIR spectra of all pure olive and adulterated oils collected at 20 °C. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Fig. 2(a) shows the NIR spectra of oleic acid measured at 20, 27, 34, and 41 °C, with the difference spectra obtained by subtracting the spectra at 27, 34, 41 °C from the spectrum at 20 °C (displayed at the top). The choppy noise-like peaks over the 7500-7100 cm^{-1} range in the difference spectra were caused by absorption of ambient water

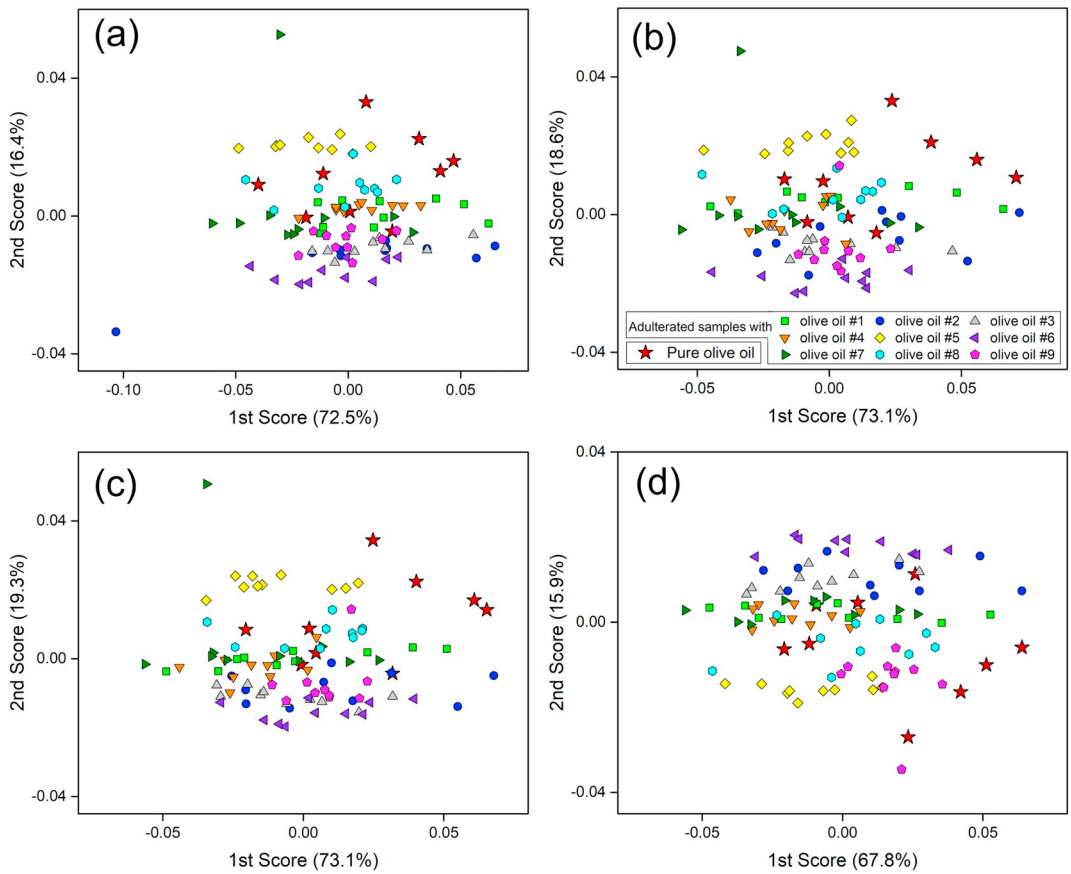

Fig. 4. Score scatter plots (1st vs. 2nd score) obtained from spectra of samples collected at 20 (a), 27 (b), 34 (c), and 41 °C (d).

vapor. The bands apparently change with temperature variation, although the magnitude of change is small. The overall absorbance decreases with temperature elevation due to the decrease of sample density, i.e., fewer molecules in a spectroscopic sample volume. Therefore, the magnitude of difference compared to the spectrum at 20 °C becomes greater with increase of temperature. Noticeably the band shape around 6910 cm^{-1} varies, while the corresponding intensity increases. Overall, the spectral variations were gradual and apparent, while the difference patterns were similar to each other without peculiar features at a certain temperature.

Fig. 2(b) and (c) show the temperature-varied NIR spectra of linoleic and linolenic acids, respectively, with the corresponding difference spectra. Similarly, the absorbance of bands decreases with temperature elevation, and band shape changes around 6910 cm^{-1} are noted. Meanwhile, the difference in spectral features among the three fatty acids is dissimilar, and the magnitude of spectral differences in the 8800-7800 cm^{-1} range is smaller for linolenic acid. The more temperature-resistant rigid structure of linolenic acid due to inclusion of more double bonds in the chain explains the decreased spectral variation. The temperature-induced spectral variations of the fatty acids are moderately different, indicating that dynamic spectral information is potentially useful for authentication of olive oil.

3.2. NIR spectral features of olive, canola, corn, and soybean oils and adulterated samples

Fig. 3(a) shows the average NIR spectra of all pure olive, canola, soybean, and corn oils used in the study, and the shades superimposed with the spectra designate the corresponding standard deviations representing sample-to-sample variation. For olive oil, the 8260 cm^{-1} band is most intense, and the absorption of 8580 cm^{-1} band corresponding to a double bond is smallest. Since the content of oleic acid in olive oil is greatest (Table 1), the spectral features of olive oil more closely resemble those of oleic acid (Fig. 1). In contrast, the intensity of the 8580 cm^{-1} band is greatest for soybean oil due to inclusion of relatively large amounts of linoleic and linolenic acids, and the absorption at 8260 cm^{-1} is the smallest. Meanwhile, these two bands of canola and corn oils are located between those of olive and soybean oils as expected. Also, the variations of the bands in the 7500-6500 cm^{-1} range are similar to those shown in Fig. 1; for example, the most intense variations occur for oleic acid as well as olive oil. Based on the observations of Fig. 3 (a), identification of a high adulteration rate will be feasible; however, decreased accuracy is expected when the adulteration rate is small, such as the 5% in this study.

Since the employed edible oils are composed of diverse fatty acids and their compositions differ as described in Table 1, the absolute concentration differences of each fatty acid by the adulteration expect to be smaller than 5%. In addition, NIR spectral features of each fatty acid are quite analogous (Fig. 1), so NIR spectral difference informing the 5%-adulteration become less distinct. Moreover, the compositions of the edible oils are varying depending on producers as confirmed by the shades in Fig. 3(a). Based on consideration of the mentioned things, the identification of 5%-adulteration will be a challenging and uneasy job.

Fig. 3(b) shows the NIR spectra of all pure olive (red) and adulterated oils (black) collected at 20 °C. The NIR bands of samples largely overlapped (refer to the inset magnifying the 8580 cm^{-1} band) due the sample-to-sample variation and only a few spectra of adulterated samples (black color) are discernible. As expected, the 5%-adulteration did not induce a sufficient spectral difference for clear discrimination. To observe the characteristics of all pure and adulterated oil samples together, PCA was performed using all the spectra and the distribution of resulting first and second principal component (PC) scores was examined, as shown in Fig. 4(a). The scores of four pure olive oils on the

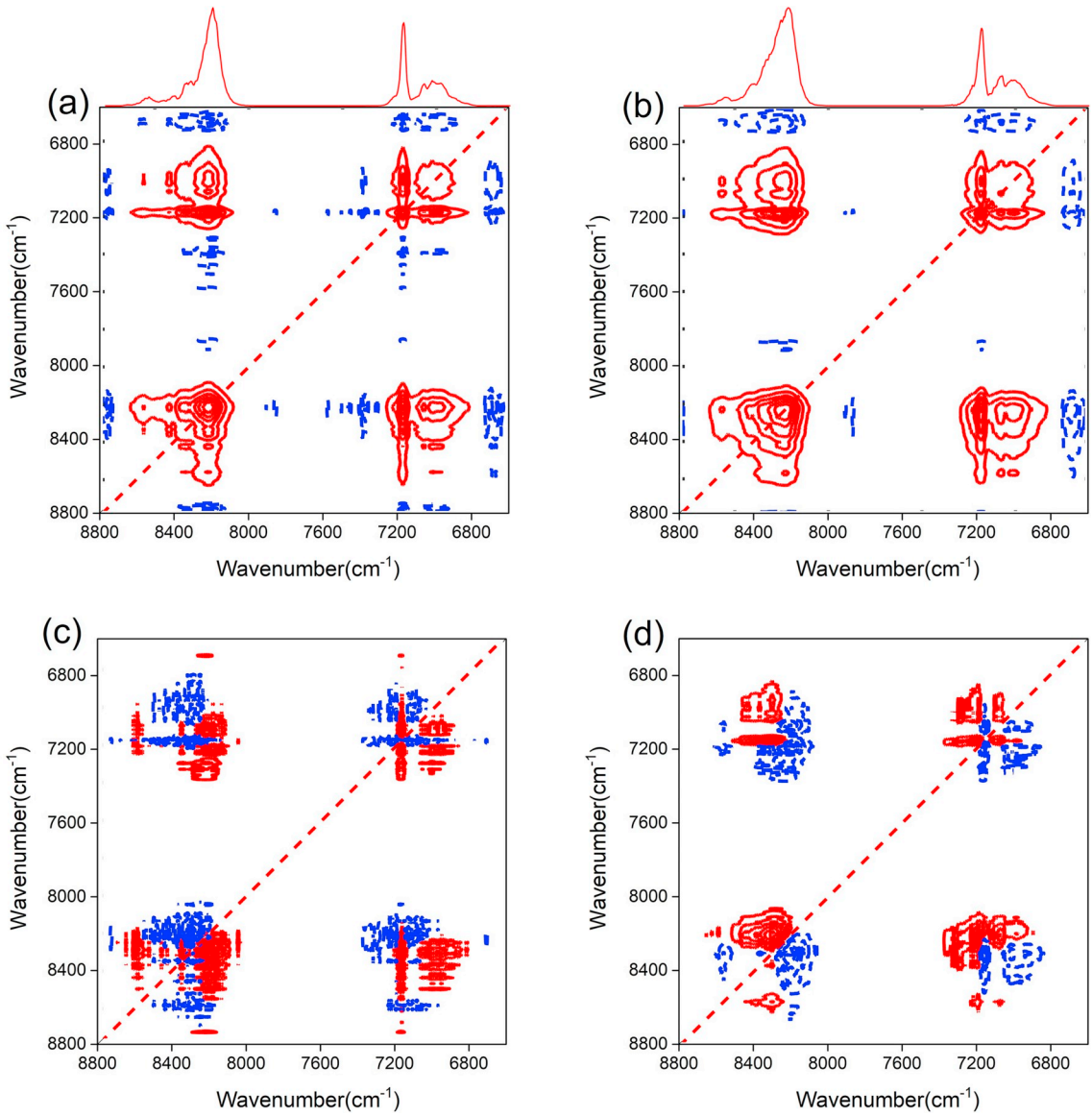

Fig. 5. 2D synchronous correlation spectra acquired from the temperature-varied (20–41 °C) spectra of pure olive oil #1 (a) and adulterated oil (#1 olive oil -#1 canola oil, (b)). The corresponding power spectra are shown on the top. The corresponding asynchronous maps are also shown at the bottom ((c) and (d)).

right (#1, #5, #8, and #9) are very near to the score group of adulterated oils without mutual overlapping. Meanwhile, the remaining five pure olive oils are within the score group of adulterated oils, indicating the compositional similarity of the samples even with the adulteration. Fig. 4(b), (c), and (d) show the score scatter plots obtained from the measurements at 27, 34, and 41 °C, respectively. The distribution patterns of the two score groups did not change noticeably by temperature variation; therefore, discrimination accuracy would not vary significantly with temperature of measurement.

3.3. 2D-COS analysis of temperature-varied sample spectra

Initially 2D-COS analysis was performed to examine whether correlation patterns of pure olive and adulterated samples under temperature variation were different or not. Fig. 5(a) and (b) show the synchronous 2D correlation spectra acquired from the temperature-varied (20–41 °C) spectra of pure olive oil (#1) and adulterated oil (#1 olive oil-#1 canola oil), respectively. The power spectra, extracted along the diagonal line in the synchronous map, are shown at the top. The detailed features in the two power spectra are somewhat dissimilar in the 8480–8300 and 7130-6930 cm^{-1} ranges, indicating different temperature-dependent behaviors. In addition, the asynchronous 2D correlation spectra of the two samples are shown in Fig. 5(c) and (d). The asynchronous 2D correlation spectra are clearly different, indicating dissimilar order of intensity changes in both sample spectra under temperature variation.

2D synchronous correlation spectra acquired from four adulterated samples (olive oil #3-canola oil #1, olive oil #6-canola oil #1, olive oil #6-soybean oil #3, and olive oil #6-corn oil #3) were further examined, as shown in Fig. 6. Comparing three synchronous 2D correlation spectra of olive oil #1-canola oil #1 (Fig. 5(b)), olive oil #3-canola oil #1 (Fig. 6(a)), and olive oil #6-canola oil #1 (Fig. 6(b)) samples, in which canola oil #1 is a common adulterant, the correlation of the olive oil #6-canola oil #1 sample is different from those of the other samples. This indicates that the thermal behavior of olive oil #6 is different from those of olive oil #1 and olive oil #3. When the three adulterated samples commonly based on olive oil #6 (olive oil #6-canola oil #1 (Fig. 6(b)), olive oil #6-soybean oil #3 (Fig. 6(c)), and olive oil #6-corn oil #3 (Fig. 6(d)) are compared, the intensity change of the band at 7186 cm^{-1} in the power spectra is sensitive to changes in both

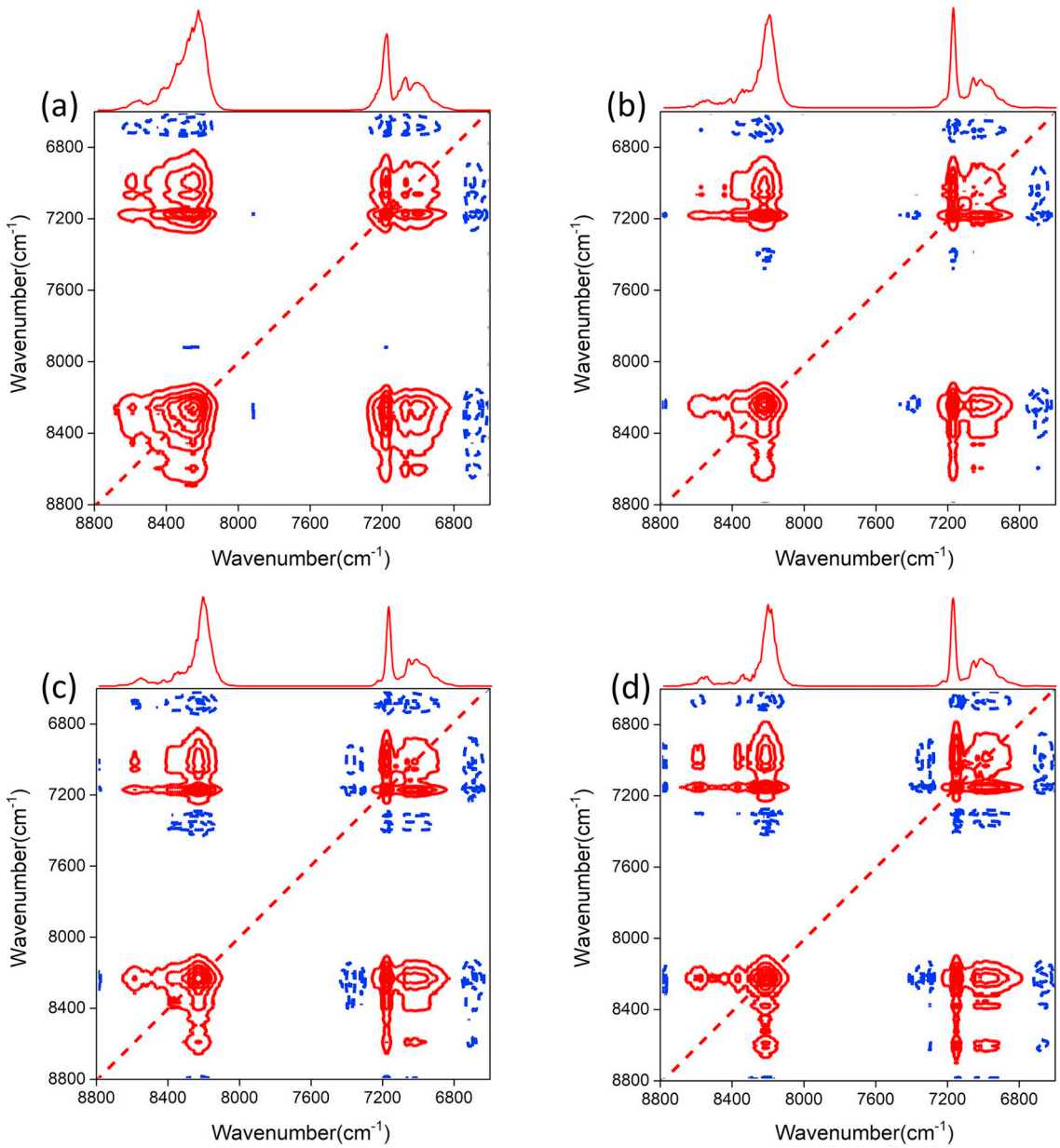

Fig. 6. 2D synchronous maps acquired from four adulterated samples (olive oil #3-canola oil #1 (a), olive oil #6-canola oil #1 (b), olive oil #6-soybean oil #3 (c), and olive oil #6-corn oil #3 (d)).

temperature and degree of unsaturation.

Fig. 7(a) shows the average power spectra of all olive and adulterated olive oils. The shade again indicates the corresponding standard deviations. The average features of pure samples are considerably different from those of adulterated samples, although the two shades are somewhat overlapped. It is clear that use of dynamic spectral information helps to enhance sample discrimination. Fig. 8(a) shows the score scatter plot generated using the power spectra. The score distribution is completely different from those of the constant-temperature measurements (Fig. 4). The differentiation of the two sample groups is dramatically enhanced by using the power spectra. Interestingly, the adulterated samples were divided into two groups, and the pure olive oil group was located between these two groups. Meanwhile, the pure olive oil group is closer to the adulterated sample group on the top. In the cases of #5, #6, #7, and #8 olive oils, the scores of pure and adulterated samples are closely located to each other. Contrastively, the scores are farther separated in the other cases (#1, #2, #3, #4, and #9 olive oils), indicative of dissimilar temperature-induced spectral variations.

In parallel, the difference power spectrum between the average power spectrum of an olive oil and that of the corresponding 10 adulterated olive oils in each case was examined (refer to Fig. 7(b)). Three difference patterns are interestingly observed here. The difference features of #5, #6, #7, and #8 olive oil cases are largely dissimilar to those of #1, #2, #3, and #4 olive oils. Also, the difference in the shape of olive oil #9 is peculiarly distinct compared to the others. Also, the magnitudes of difference for olive oils #1, #2, #3, #4, and #9 are larger, thereby explaining the larger score separation between pure and adulterated sample groups shown in Fig. 8 (a). Again, among the three cases of olive oil #1-canola oil #1 (Fig. 5(b)), olive oil #3-canola oil #1 (Fig. 6(a)), and olive oil #6-canola oil #1 (Fig. 6(b)), the 2D-COS feature of olive oil #6-canola oil #1 is relatively more similar to that of #1 olive oil ((Fig. 5(a)). This similarity explains why the corresponding scores are closely located (Fig. 8(a)).

For more detailed investigation on the distribution of scores, the corresponding loadings were examined as shown in Fig. 8 (b). The

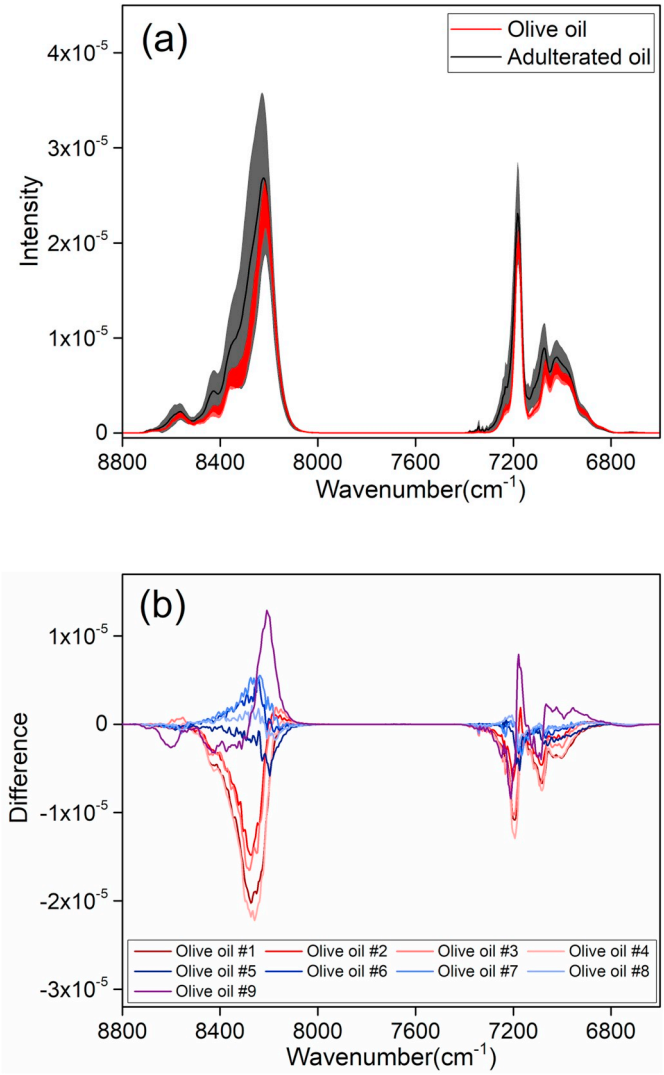

Fig. 7. Average power spectra of olive oil and adulterated olive oil samples obtained from 2D-COS analysis (a). The shades superimposed with the spectra indicate the corresponding standard deviations. Difference power spectra between the average power spectrum of an olive and that of the corresponding 10 adulterated olive oils in each case is also shown (b). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

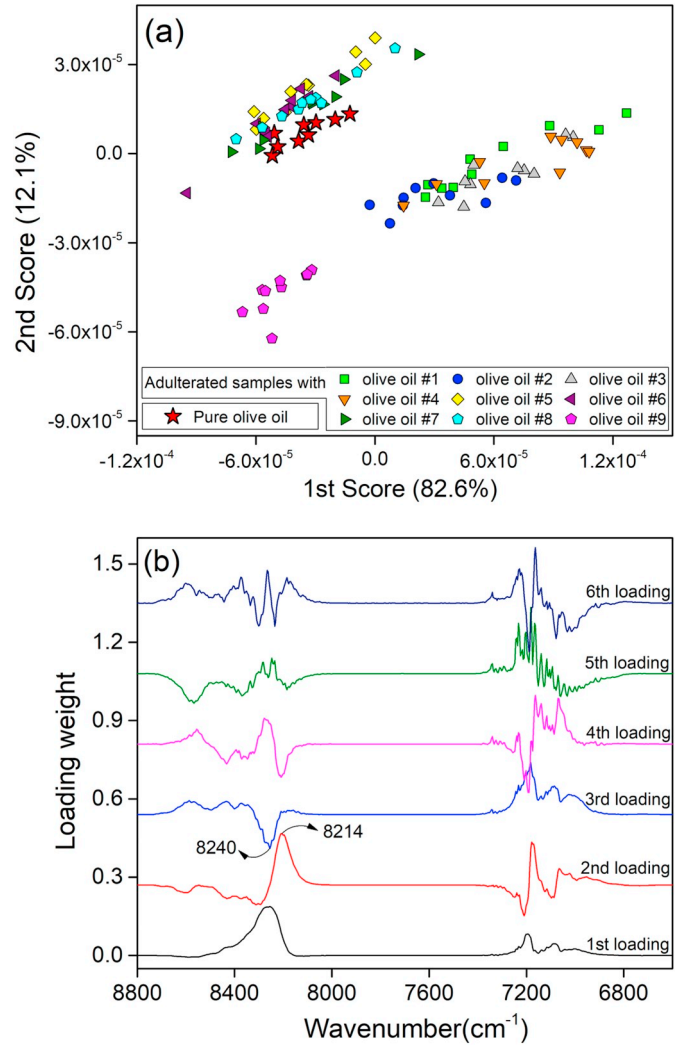

Fig. 8. Score scatter plot generated using the power spectra of all samples (a) and 1st through 6th loadings used in the PCA for generation of the score scatter plot (b).

variation of power spectra among the samples is substantial (Fig. 7 (a)) and the 1st loading is constructed to handle this spectral variation. So, its feature is rather broad, and discrimination of the two groups in the first score domain is indistinct. Then, the 2nd loading weights the 8214 cm^{-1} peak corresponding to the position of the maximum peak in the average pure olive oil power spectrum, and the subsequent discrimination improves in the 2nd score domain, as shown in Fig. 8 (a). The 3rd loading emphasizes the 8240 cm^{-1} peak, which is very near the maximum peak in the average adulterated oil power spectrum. The 4th, 5th, and 6th loadings additionally describe the features of the power spectra by weighting on different locations under the main peaks. Overall, a clear explanation of the larger score separation in the five cases (olive oils #1, #2, #3, #4, and #9) would be difficult. Nonetheless, it is probable that the molecular interactions of components in these olive oils with those of the adulterating oils are more temperature-susceptible, which is effectively recognized by 2D-COS analysis.

3.4. Evaluation of discrimination accuracy using one-class k-NN

Since the scores of adulterated samples obtained from their power spectra were separated into two groups, a method able to classify a sample regardless of distribution pattern would be desirable. For this purpose, one-class *k*-NN described in the experimental section was employed. The first six PC scores describing 99.0% of the total spectral variation in the power spectra were used for one-class *k*-NN. The same procedure was performed for the constant-temperature measurements.

Meanwhile, the sample set used in this study contains the unbalanced number of pure and adulterated samples, which is 9 and 90 samples, respectively. Under this situation, final accuracy could be excessively governed by identification accuracy of adulterated samples. For example, if all the 99 samples are predicted as adulterated samples, resulting accuracy is still high (90.9%) although all the 9 pure samples are incorrectly identified. This is a typical example showing misleading interpretation of accuracy in the case of using unbalanced dataset.

To assess discrimination accuracy with avoiding the mentioned problem, out of 9 pure samples, 6 and 3 samples were initially assigned into a training and validation set, respectively. To make a validation set balanced, only 3 adulterated samples were randomly selected from the total of 90 adulterated samples and added into a first validation set. So, this validation set contains 3 pure and 3 adulterated samples. Then, another 3 adulterated samples were randomly selected again out of the

Table 2
Accuracy, sensitivity and specificity of discrimination between pure olive oils and adulterated samples using NIR spectra measured at each temperature or power spectra obtained from 2D-COS analysis using one-class *k*-NN.

Measurement temperature	Number of PCs	[a]Accuracy	[b]Sensitivity	[c]Specificity
20 °C (NIR)	6	51.9%	74.0%	29.8%
27 °C (NIR)	6	53.7%	74.7%	32.8%
34 °C (NIR)	6	50.9%	69.3%	32.4%
41 °C (NIR)	6	55.8%	82.0%	29.6%
20–41 °C (Power)	6	86.4%	79.3%	93.4%

[a] Calculated by (TP + TN)/(TP + FN + FP + TN)*100.
[b] Calculated by (TP)/(TP + FN)*100.
[c] Calculated by (TN)/(FP + TN)*100.

remained 87 adulterated samples for a second validation set. This procedure repeated 30 times until all the 90 adulterated samples were predicted without duplicated sampling. Also, during the 30 time-validation, 3 pure samples were also randomly selected in each step out of the total 9 pure samples.

Table 2 shows the accuracy, sensitivity, specificity resulted from one-class *k*-NN with the 30 time-validation. TP, FP, TN, and FN stand for the numbers of true positive, false positive, true negative, and false negative samples. The use of 2 nearest neighbors to determine both *K* and *J* samples provided the optimal result. At a glance, the accuracies acquired in the four different constant-temperature measurements range from 50.9 to 55.8%, substantially inferior to that obtained in the varying-temperature measurement (86.4%). Especially, the specificities in the constant-temperature measurements are low (29.6–32.8%). This means that many adulterated samples are incorrectly identified as pure samples. The inferior accuracies of the constant-temperature measurements are easily expectable since the scores of five pure olive oils are inseparable and located within the score group of adulterated oils (Fig. 4). In contrast, both groups are quite separable in the score domain constructed using the power spectra (Fig. 8); therefore, the accuracy improves to 86.4% and significant enhancement in the specificity is notable. Overall, the use of power spectra effectively reflecting dynamic temperature-induced spectral changes by 2D-COS analysis was useful to improve discrimination accuracy in comparison with employing NIR spectra acquired at constant temperature.

4. Conclusions

The use of power spectra from 2D-COS analysis reflecting dynamic temperature-induced spectral variation of sample components was more informative to enhance sample discrimination and identify olive oil adulteration more accurately in this study. When olive oils were adulterated with other edible oils, the features of power spectra from 2D-COS analysis were more distinct compared with those of spectra collected at constant temperature. So, the improvement of discrimination accuracy was then consequential. We think that the proposed strategy is beneficial to improve accuracy of discriminant analysis for various samples since temperature-perturbed spectral variation is more sample-descriptive, and also execution of the experiment is simple by using a temperature-controllable spectroscopic cell. Further, it is readily expandable for Raman and IR spectroscopic discriminant analysis of samples, since temperature-induced Raman and IR spectral variations are individually characteristic and diverse.

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